Whole-Body Safety Filtering of Flow-Matching Policies with Geometrically Refined Distance Fields

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***Abstract*—Flow Matching (FM) policies learn multimodal manipulation behaviour from a few dozen demonstrations, but a policy trained without obstacles has no notion of one and will drive straight through it. Constraining the sampling process or retraining on cluttered scenes pays for safety with distribution shift or with data. We filter execution instead: a frozen FM policy emits nominal trajectories at 3 to 5 Hz while a whole-body Control Barrier Function shield projects the velocity command at 100 Hz, backed by a 1 kHz reflex brake in the real-time controller. The enabling component is a robot distance field accurate in its *gradient*: nine Bernstein tensor polynomials refined against 20 000 verified surface normal rays under Eikonal and directional losses, reaching 1.96 mm mean distance error and 5.4° median gradient direction error against exact meshes. We derive the exact closed-form joint-space gradient, which matches automatic differentiation to single-precision roundoff while running up to $1.75\times$ faster, and a monotone out-of-domain minorant that never inverts the correction direction. Every gap between the continuous-time certificate and sampled execution becomes an online-monitored assumption: a 26-trajectory campaign calibrates the tracking bound at 97.9 % coverage and shows a static worst-case bound would be 3.7 to 6.9 times larger.**

***Index Terms*—Collision avoidance, safety filtering, control barrier functions, learning from demonstration, distance fields.**

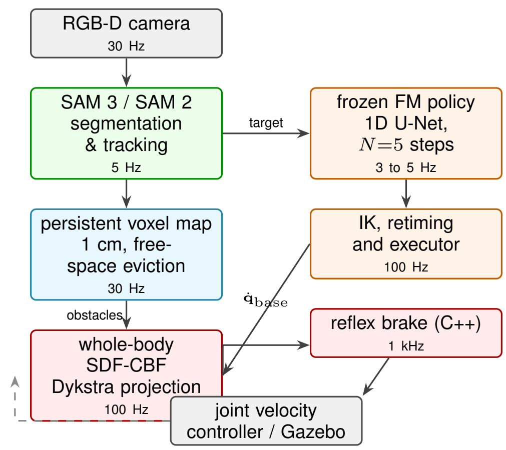

Fig. 1: Asynchronous pipeline. Task intent and safety are fully decoupled: the frozen FM policy never sees the obstacle map, and the shield never waits for the planner. A stale conditioning cloud degrades the nominal trajectory without ever suspending the 100 Hz command.

I. Introduction

GENERATIVE behaviour cloning has changed how manipulation policies are built. Diffusion policies [1] and Flow Matching (FM) [2], [3] recover the multimodality of human demonstrations instead of averaging it away, and FM adds a property that matters for control: because its conditional paths are close to straight, a trajectory segment is sampled in a handful of network evaluations, which puts closed-loop replanning within reach where diffusion remains too slow.

A generative policy is nonetheless an exact reflection of its data. Trained in a clear workspace, it has never met an obstacle and holds no mechanism for one: placed in front of a bowl or a hand absent from its training set, it produces a trajectory that goes through them. Two families of answers exist and both charge a price. Constraining the sampling process itself [4] pushes the model through states it never saw at training time, shifting the distribution and loading every integration step. Teaching avoidance to the model demands demonstrations enriched with obstacles far beyond the few dozen that suffice for the nominal task, repeated for every family of scenes; having an operator correct failures at deployment [5] displaces that effort rather than removing it.

The third answer filters execution rather than generation: the policy stays frozen and a safety layer corrects the command sent to the robot [6], [7]. This preserves both the learned intent and a formal guarantee, but its deployment runs into a rarely addressed prerequisite. Protecting an articulated robot requires a signed distance between its *whole body* and its environment, queried together with its gradient, for thousands of measured points at control rate. Without it, existing implementations restrict themselves to the end effector, leaving the elbow and forearm free to hit what the gripper avoids, or assume obstacles described by primitives known in advance, which no sensed scene satisfies.

We present a system in which intent and safety are decoupled end to end (Fig. 1), and, more importantly, in which every gap between the textbook certificate and sampled execution is named, tightened, and monitored online. Our contributions are:

- A two-stage offline scheme for Bernstein robot distance fields that refines a value regression against 20 000 *verified* normal rays under Eikonal and directional-alignment losses, yielding a field whose gradient, not only whose value, is usable by a CBF (Sec. IV).
- An exact closed-form joint-space distance gradient obtained from link geometric Jacobians, whose cost is independent of the number of joints and which matches automatic differentiation to single-precision roundoff while

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running faster (Sec. IV-D, Sec. VI-C).
- A monotone out-of-domain minorant $\hat{d} = \sqrt{d(\mathbf{c})^2 + \|\bar{\mathbf{x}} - \mathbf{c}\|^2}$ that under-estimates the true distance while preserving the outward gradient orientation, where the usual Lipschitz extension inverts it (Sec. IV-C).
- A shield carrying three additive tightenings, for sampled-data hold, tracking perturbation and environment motion, each paired with a telemetry channel that turns it into a monitored assumption rather than a postulate, plus a distribution-free calibration of the tracking bound (Sec. V, VI-F).
- An honest treatment of the Lipschitz constants the sampled-data certificate consumes, including a valid analytic Bernstein-coefficient bound that we report and then reject as unusable, being about forty times the worst sampled case (Sec. VI-D).

II. Related Work

Generative policies and safety. Behaviour cloning through generative modelling [1] captures multimodal action distributions from small demonstration sets, and rectified flows [2], [3] make few-step deterministic sampling practical. Safety is then sought either inside generation, through guided or constrained sampling [4], or outside it, through a downstream filter [6], [7]. We take the second route and, unlike [6], enforce the constraint on the whole kinematic chain against a sensed, non-parametric obstacle set.

Distance fields for manipulators. Mapping frameworks [8] model the environment, but querying a robot against them at control rate remains the bottleneck. Robot Distance Fields [9] represent each link SDF by a Bernstein tensor product in a normalised link frame, and [10] uses them for reactive avoidance under task priority. These works fit *values* by recursive least squares and obtain joint gradients by automatic differentiation or finite differences. Value-only fitting leaves the spatial gradient noisy in direction and off unit norm exactly where a CBF needs it, which is the gap our refinement stage closes.

Control barrier functions in practice. CBFs [11] give forward invariance of $\mathcal{C} = \{h \geq 0\}$ and, with a quadratic program, minimally alter a nominal input. Sampled deployment requires sampled-data treatments [12], input-to-state safety for actuation error [13], and multi-rate nesting of a cheap high-rate certificate under a slower solver [14]. We combine these three on a learned whole-body barrier and, rather than inflating a static margin, calibrate and monitor each at runtime.

III. System Overview and Problem Statement

A. Asynchronous tiers

Four tiers run at their own rate and communicate by message, never blocking downstream (Fig. 1). *Perception* grounds the target from a text prompt with SAM 3 [15] and propagates the mask at 5 Hz with SAM 2 [16]; pinhole back-projection and forward kinematics yield a target cloud and an obstacle cloud. *Mapping* accumulates obstacles in a sparse 1 cm voxel grid. Outside the current field of view a voxel persists indefinitely, which is precisely what protects the arm from obstacles it can no longer see; inside it, a GPU free-space eviction re-projects each voxel along its pixel ray and discards it as soon as the measured depth exceeds its own by one voxel, so a removed obstacle disappears with no ageing delay. *Generation* is a 1D conditional U-Net [1] FM policy predicting $H = 16$ tool poses (position, 6D rotation [17], gripper state) conditioned on a 1024-point cloud encoded as in [18], integrated with $N = 5$ Euler steps and replanned at 3 to 5 Hz. *Reactive control* solves the shield at 100 Hz and re-certifies the held command at 1 kHz inside `ros_control`. No edge in Fig. 1 carries obstacles into the planner, which is what makes the policy replaceable without touching the certificate.

B. Problem statement

Let $\mathbf{q} \in \mathbb{R}^7$ be the measured configuration and $\dot{\mathbf{q}}_{\mathrm{base}}$ the nominal joint velocity produced by the frozen policy after inverse kinematics, retiming and tracking feedback. For each of the nine protected links, namely links 2 to 7, the hand and the two fingers, let $h_l(\mathbf{q})$ be a barrier built from the whole-body distance field against the obstacle set $\mathcal{P}_{\mathrm{obs}}$. We seek the admissible command closest to the nominal one in a task metric $\mathbf{W} \succ 0$:

$$\dot{\mathbf{q}}^\star = \underset{\dot{\mathbf{q}} \in \mathbb{R}^7}{\arg\min}\ \tfrac{1}{2}\|\dot{\mathbf{q}} - \dot{\mathbf{q}}_{\mathrm{base}}\|^2_{\mathbf{W}} \quad \text{s.t.} \quad \nabla h_l^\top \dot{\mathbf{q}} \ \geq\ r_l,\ \forall l \in \mathcal{A}, \tag{1}$$

where $\mathcal{A}$ is the active set and r_l the tightened right-hand side of Sec. V-C. Safety lives in the constraints and intent lives in the cost, so no nominal command, however strong, can violate a safety half-space.

IV. Geometrically Refined Bernstein Robot Distance Field

A. Representation

Following [9], each protected link mesh is centred on its bounding-box centre $\mathbf{o}_l$ and scaled by $s_l = \max_{\mathbf{v} \in \mathcal{V}_l} \|\mathbf{v} - \mathbf{o}_l\|$, so that $\bar{\mathbf{x}} = (\mathbf{x} - \mathbf{o}_l)/s_l$ lies in $[-1, 1]^3$. On that cube the normalised field is a tensor product of Bernstein polynomials,

$$\bar{d}_l(\bar{\mathbf{x}}) = \sum_{a,b,c=0}^{n-1} w^{(l)}_{abc} B_a(u_x) B_b(u_y) B_c(u_z) = \mathbf{\Phi}(\bar{\mathbf{x}})\, \mathbf{w}_l, \tag{2}$$

with $\mathbf{u} = (\bar{\mathbf{x}} + \mathbf{1})/2$ and $n = 12$ basis functions per axis, that is 1728 coefficients per link; the metric distance is $d_l = s_l \bar{d}_l$. The representation is $\mathcal{C}^\infty$ with an exact analytic gradient, linear in its parameters, and parallelises on GPU.

B. Two-stage training

Stage one reproduces the reference protocol [9]: exact signed-distance labels are sampled near the surface and uniformly in the cube, and since (2) is linear in $\mathbf{w}_l$ the weights are fitted by ridge-regularised recursive least squares. This yields an accurate *value* and says nothing about the gradient, whose direction stays noisy near the surface and whose norm drifts from unity, while the shield consumes exactly that gradient: its direction sets the correction direction in (1) and its norm the amplitude.

Stage two, the contribution here, refines those weights on a dedicated supervision set of 20 000 *verified* normal rays per link. A ray is built by displacing a surface point along its facet normal by an offset δ, with 75 % of the offsets drawn in the critical shell $[1, 10]$ mm consistent with $d_{\text{safe}} = 15$ mm. A displaced point does not necessarily keep its anchor as nearest neighbour, so each candidate is checked against the exact mesh and kept only if its recomputed exact distance matches δ within 0.25 mm and the exact gradient direction lies within 5° of the anchor normal. This rejects rays corrupted by the medial axis and by local curvature, which would otherwise inject a wrong distance and a wrong direction at once. The weights are then optimised with Adam on

$$\begin{aligned}\mathcal{L} = \mathcal{L}_{\text{near}} + \mathcal{L}_{\text{unif}} + \lambda_{\text{ray}}\mathcal{L}_{\text{ray}} + \lambda_{\text{eik}}\,\mathbb{E}\big[(\|\nabla\bar{d}\| - 1)^2\big] \\ + \mathbb{E}\Big[\omega\big(\lambda_{\text{dir}}(1 - \widehat{\nabla\bar{d}}\cdot\hat{\mathbf{n}}) + \lambda_{\text{vec}}\|\nabla\bar{d} - \hat{\mathbf{n}}\|^2\big)\Big],\end{aligned} \tag{3}$$

where the first three terms preserve value accuracy, the Eikonal term pushes $\|\nabla\bar{d}\|$ towards one, and the last two align the gradient with the exact outward direction under a proximity weight $\omega = 1 + 4e^{-\bar{d}_{\text{exact}}/\bar{\delta}_c}$ that concentrates the effort within roughly d_{safe} of the surface. The Eikonal term is a sampled quadratic penalty, not a constraint: the resulting Eikonal property is a measured property and is reported as such in Sec. VI-B.

C. Monotone out-of-domain minorant

Queries fall outside $[-1, 1]^3$ whenever an obstacle is far from a link. Clamping to $\mathbf{c} = \text{clamp}(\bar{\mathbf{x}}, -1, 1)$ and applying the usual Lipschitz bound $d(\mathbf{c}) - \|\bar{\mathbf{x}} - \mathbf{c}\|$ produces a minorant that *decreases* with the excursion, so its gradient points back towards the link and the shield pushes the arm the wrong way. We use instead

$$\hat{d}(\bar{\mathbf{x}}) = \sqrt{d(\mathbf{c})^2 + \|\bar{\mathbf{x}} - \mathbf{c}\|^2}. \tag{4}$$

Every surface point $\mathbf{s}$ lies inside the cube, so along each clamped axis j we have $|x_j - s_j| = |c_j - s_j| + |x_j - c_j|$, hence $\|\bar{\mathbf{x}} - \mathbf{s}\|^2 \geq \|\mathbf{c} - \mathbf{s}\|^2 + \|\bar{\mathbf{x}} - \mathbf{c}\|^2$ for every $\mathbf{s}$ and therefore $\hat{d} \leq d$ always: the extension under-estimates the true distance, and being monotone in the excursion the correction direction never inverts. Its gradient,

$$\nabla\hat{d} = \frac{d(\mathbf{c})\,\nabla d(\mathbf{c})|_{\text{int}} + (\bar{\mathbf{x}} - \mathbf{c})}{\hat{d}}, \tag{5}$$

joins the polynomial continuously at the boundary; the join is differentiable piecewise but not continuously differentiable, a limitation we return to in Sec. VI-D.

D. Exact closed-form joint-space gradient

Let $\mathbf{R}_l^b(\mathbf{q})$ and $\mathbf{d}_l^b(\mathbf{q})$ be the pose of link l, and $\mathbf{x}_{l,i} = (\mathbf{R}_l^b)^\top(\mathbf{p}_i - \mathbf{d}_l^b)$ the obstacle point in the link frame. The obstacle is fixed in the base frame but appears to move in the link frame; differentiating the closure relation $\mathbf{p}_i = \mathbf{d}_l^b + \mathbf{R}_l^b\mathbf{x}_{l,i}$ with $\dot{\mathbf{p}}_i = \mathbf{0}$, substituting the linear and angular link Jacobians $\dot{\mathbf{d}}_l^b = \mathbf{J}_{L,l}^b\dot{\mathbf{q}}$ and $\dot{\mathbf{R}}_l^b = [\mathbf{J}_{A,l}^b\dot{\mathbf{q}}]_\times\mathbf{R}_l^b$, and applying the cross-product identities gives

$$\mathbf{J}_{x_{l,i}} = -(\mathbf{R}_l^b)^\top\mathbf{J}_{L,l}^b + [\mathbf{x}_{l,i}]_\times(\mathbf{R}_l^b)^\top\mathbf{J}_{A,l}^b. \tag{6}$$

Composing with the metric spatial gradient expressed in the base frame, $\mathbf{g}_{l,i} = \mathbf{R}_l^b\nabla_{\mathbf{x}}d_{l,i}$, and writing $\mathbf{r}_{l,i} = \mathbf{p}_i - \mathbf{d}_l^b$, the affine normalisation and the metric rescaling cancel and the joint-space gradient collapses to

$$\boxed{\nabla_{\mathbf{q}}d_{l,i} = -(\mathbf{J}_{L,l}^b)^\top\mathbf{g}_{l,i} - (\mathbf{J}_{A,l}^b)^\top\big(\mathbf{r}_{l,i} \times \mathbf{g}_{l,i}\big).} \tag{7}$$

Evaluating (7) costs two matrix-vector products and one cross product per obstacle point, is independent of the number of joints, and involves no numerical differentiation step. Its exactness is verified against automatic differentiation in Sec. VI-C.

V. Whole-Body Reactive SDF-CBF Shield

A. Barrier assembly

The whole-body field is evaluated over the map and the $K = 64$ points of smallest distance are kept. Forward kinematics, field evaluation and top-K selection are fused into a single fixed-shape CUDA graph, padded with distant dummy points so the population N stays constant, as the constant-selection assumption of Sec. V-G requires. The selection is recomputed every 100 Hz cycle, so the active set carries at most 10 ms of staleness, a delay budgeted into d_{safe}.

Per-link distances are aggregated by a numerically stabilised soft-min of temperature α,

$$h_l^{\text{soft}} = m_l - \alpha\log\Big[\sum_{i=1}^{N} e^{-(d_{l,i} - m_l)/\alpha}\Big] - d_{\text{safe}}, \quad m_l = \min_i d_{l,i}, \tag{8}$$

whose gradient is the convex combination $\nabla_{\mathbf{q}}h_l = \sum_i \omega_i\nabla_{\mathbf{q}}d_{l,i}$ with $\omega_i = \text{softmax}_i(-d_{l,i}/\alpha)$. The soft-min under-estimates the true minimum by at most $\alpha\log N$, which is safe but costs usable workspace: the arm stops further out than the geometry requires and an admissible path through a tight passage can be refused. The deployed filter therefore decouples the two roles, sending the *exact* worst-case value $h_{\min,l} = \min_i d_{l,i} - d_{\text{safe}}$ to the right-hand side while keeping the soft-min gradient, whose weights act as a smooth selector preserving spatial continuity when the dominant obstacle switches; the consequence for the certificate is stated in Sec. V-G. A grasped target is protected as an extra link with its own partition of the selected points, so N in (8) is the population of the relevant group. The $K_a = 9$ links of smallest h_l are retained as distinct constraints rather than collapsed into a single minimum, which would leave every other link free to advance towards its own obstacle and would switch abruptly as soon as a second link became critical.

B. Saturated class-$\mathcal{K}$ right-hand side

The textbook condition $\nabla h^\top\dot{\mathbf{q}} \geq -\kappa h$ has two defects for a perceived barrier: far from obstacles it authorises arbitrarily large approach speeds, and under violation it demands a recovery speed proportional to penetration, which converts perception noise on h into violent motor jerks. We saturate both regimes inside the activation band $h \leq h_{\text{act}}$:

$$b(h) = \begin{cases} -v_{\text{in}}\min(h/h_{\text{act}}, 1), & h > 0, \\ \;\;v_{\text{rec}}\min(-h/d_{\text{rec}}, 1), & h \leq 0. \end{cases} \tag{9}$$

Setting $\gamma(h) = -b(h)$ recovers $\nabla h^\top\dot{\mathbf{q}} \geq -\gamma(h)$. Restricted to the operating band $[-d_{\text{rec}}, h_{\text{act}}]$, γ vanishes at zero, is strictly increasing and locally Lipschitz, hence extended class-$\mathcal{K}$ near the boundary, which is what the invariance theorem of [11] requires; beyond that band the saturation adds two plateaus far from the boundary without altering the guarantee. With the deployed values the plateaus have equal amplitude but recovery is more than five times steeper than approach, 33.3 against 6.25 s^{-1}.

C. Three tightenings, each a monitored assumption

Only γ must be class-$\mathcal{K}$. The three terms below are added to $b(h)$ and are non-negative by construction, so they strengthen the constraint without changing the dynamics of h at the boundary and need not themselves be class-$\mathcal{K}$.

Environment motion. The classical right-hand side bounds only the robot's own contribution to $\dot{h}_l$, while an approaching obstacle consumes margin outside the robot's control. We estimate that contribution online and model-free, as the residual $\Delta h_l/\Delta t - \nabla h_l^\top \dot{\mathbf{q}}$ between the measured total variation and the robot-induced part, smoothed by a first-order filter of constant τ_{env} and reset on obstacle switching or recession. Each active half-space is tightened by

$$\eta_l = \mathrm{clip}\big(-\hat{\dot{d}}_l - \varepsilon_{\mathrm{env}},\, 0,\, \eta_{\max}\big), \tag{10}$$

so the robot yields in proportion to the approach speed. The dead zone $\varepsilon_{\mathrm{env}}$ keeps $\eta_l = 0$ on static scenes and $\eta_{\max}$ avoids demanding unrealistic accelerations. Being per link, the estimate catches an obstacle approaching the elbow even when the end effector is still. This term is a robustification and we claim no formal guarantee against moving obstacles.

Tracking perturbation. The low-level controller realises the commanded velocity only approximately, $\dot{\mathbf{q}} = \dot{\mathbf{q}}_{\mathrm{cmd}} + \boldsymbol{\delta}$. Following input-to-state safety [13] we tighten by the worst alignment of that perturbation with the gradient,

$$\nabla h_l^\top \dot{\mathbf{q}}_{\mathrm{cmd}} \;\geq\; b(h_l) + \eta_l + \varepsilon_{\mathrm{p}} \|\nabla h_l\|, \tag{11}$$

which restores the original condition for the realised velocity as soon as $-\nabla h_l^\top \boldsymbol{\delta}/\|\nabla h_l\| \leq \varepsilon_{\mathrm{p}}$. Bounding the full norm $\|\boldsymbol{\delta}\|$ instead would assume $\boldsymbol{\delta}$ exactly anti-aligned with ∇h_l at every cycle, which overloads the certificate without strengthening it. Rather than a pessimistic theoretical majorant, ε_{p} is calibrated on the measured distribution (Sec. VI-F) and enforced as a monitored assumption: the projected perturbation is recomputed every controller tick and any excursion triggers the reflex veto below.

Sampled-data hold. Between two solves the command is held and the state travels a straight segment, so following [12] each active half-space is tightened by

$$\mu_l(\dot{\mathbf{q}}) = \tfrac{1}{2} L_l \|\dot{\mathbf{q}}\|^2 T, \tag{12}$$

with L_l the per-link constant of Sec. VI-D and $T = 10$ ms the nominal period, entering as an upper bound so that any shorter effective period makes the constraint more conservative, never more permissive. Since (12) is quadratic in $\dot{\mathbf{q}}$, embedding it directly would turn the QP into a QCQP, and replacing it by its tangent would relax a convex right-hand side and is unsafe; it is therefore evaluated at a velocity envelope v_{env} upper bounding the realised speed over the hold. The recoil is proportional to the square of the speed and vanishes at rest.

The constraint actually solved is therefore

$$\nabla h_l^\top \dot{\mathbf{q}}_{\mathrm{cmd}} \;\geq\; r_l \;\equiv\; b(h_l) + \eta_l + \varepsilon_{\mathrm{p}} \|\nabla h_l\| + \mu_l, \tag{13}$$

and since all three additions are non-negative, $r_l \geq -\gamma(h_l)$, so (13) implies the standard CBF condition and preserves the invariance certificate.

D. Task-metric projection

With $\mathbf{W} = \mathbf{I}$ the correction follows $\nabla_{\mathbf{q}} h_l$ directly, loading the joints nearest the base because they have the largest kinematic lever, so avoiding an obstacle with the elbow throws the end effector off task. We instead measure the cost of a joint velocity by the Cartesian displacement it inflicts on the controlled link,

$$\mathbf{W} = \mathbf{J}_p^\top \mathbf{J}_p + w_{\mathrm{rot}} \mathbf{J}_o^\top \mathbf{J}_o + \lambda_{\mathbf{W}} \mathbf{I}, \tag{14}$$

so the correction is oriented along $\mathbf{W}^{-1} \nabla h_l$ rather than the gradient itself and lands preferentially in the task nullspace. The metric changes the distribution of the deviation, never the admissible set, so safety is untouched.

Several simultaneously active half-spaces admit no closed-form projection, so (1) is solved by Dykstra's variant of cyclic projections [19]. Unlike plain cyclic projections, which accumulate conservatism, Dykstra stores a correction vector per active constraint and re-injects it at each sweep, converging to the exact $\mathbf{W}$-projection independently of sweep order. Its fixed iteration count and branch-free structure are what allow capture as a CUDA graph; a generic QP solver would be exact but its data-dependent branching is not capturable. The loop is truncated at $N_{\mathrm{it}} = 6$ sweeps.

Output smoothing, joint-limit saturation and sweep truncation all act after the solver and can each re-introduce a violation. The smoothing is therefore folded into the QP objective rather than applied downstream, and the constraint is re-evaluated on the saturated velocity with the same r_l; any residual violation is corrected by a capped repair step in the same metric $\mathbf{W}$. Lines whose gradient norm collapses below $g_{\min}$ are dropped rather than attenuated, since below that threshold even the maximum joint velocity yields under 2 cm/s of clearance and the direction is dominated by noise. Admissibility of the *published* command is finally re-measured every cycle and exported as `constr_final`: it is observed, not assumed.

E. 1 kHz reflex brake

The QP certificate is instantaneous, and between two solves the command is held. Instead of enlarging d_{safe} conservatively to absorb that hold, a second layer re-certifies the held command in C++ inside the 1 kHz `ros_control` loop, consuming the atomic command-plus-certificate packet published at 100 Hz. It scales the command by a single scalar,

$$\dot{\mathbf{q}}_{\mathrm{exec}} = \alpha\, \dot{\mathbf{q}}_{\mathrm{final}}, \qquad \alpha \in [0, 1], \tag{15}$$

acting as a purely kinematic brake that modulates speed without ever deviating direction. With ∇h_l assumed L_l-Lipschitz in $\mathbf{q}$, the quadratic lower bound

$$h(\mathbf{q}_0 + \Delta\mathbf{q}(t)) \;\geq\; h(\mathbf{q}_0) + \nabla h^\top \Delta\mathbf{q}(t) - \tfrac{L}{2} \|\Delta\mathbf{q}(t)\|^2 \tag{16}$$

is concave in t, so checking it at $t = T_{\mathrm{rfx}}$ certifies the whole held segment. Rather than requiring $h \geq 0$, the brake certifies an admissible decay,

$$h\big(\mathbf{q}_0 + \Delta\mathbf{q}(T_{\mathrm{rfx}})\big) \;\geq\; h(\mathbf{q}_0)\big(1 - \kappa_{\mathrm{rfx}} T_{\mathrm{rfx}}\big), \tag{17}$$

TABLE I: Deployed shield parameters.

Symbol	Description	Value
d_{safe}	safety margin on the learned field	15 mm
α	soft-min temperature	0.002
K, K_a	critical points / constrained links	64, 9
h_{act}, d_{rec}	activation band / recovery depth	80, 15 mm
$v_{\text{in}}, v_{\text{rec}}$	approach / recovery saturation on $\dot{h}$	0.50 m/s
ε_{p}, t_{veto}	monitored tracking bound / veto hold	0.025 rad/s, 50 ms
$\varepsilon_{\text{env}}, \eta_{\max}$	dead zone / cap of η_l	0.05, 0.5 m/s
$g_{\min}$, N_{it}	gradient threshold / Dykstra sweeps	0.03, 6
$w_{\text{rot}}, \lambda_{\mathbf{W}}$	task metric weights	0.01, 0.01
$f_{\text{rfx}}, \kappa_{\text{rfx}}$	reflex rate / decay slope	1 kHz, 25 s^{-1}
$k_{\text{clear}}, h_{\text{ns}}$	postural clearance gain / band	0.3 rad/s, 50 mm
$\tau_{\text{env}}, \tau_{\text{safe}}, \tau_{\text{rel}}$	filter constants	0.05, 0.1, 0.05 s

a linear class-$\mathcal{K}$ condition that nests under the solver's [14]. The brake tolerates a steeper instantaneous decay ($\kappa_{\text{rfx}} = 25\ \text{s}^{-1}$) than the solver's macroscopic slope ($6.25\ \text{s}^{-1}$), and that gap is exactly the reserve that absorbs inter-sample uncertainty without needlessly rejecting valid commands. Each active line yields a quadratic inequality in α; the brake returns the largest α common to all lines and stops ($\alpha = 0$) if their intervals do not intersect. T_{rfx} is the time actually elapsed since the packet was published, so solver latency ages the state rather than shortening the horizon. Engagement is instantaneous and release is smoothed by τ_{rel}: braking late is a safety error, releasing late only a performance cost. Lines already in violation are certified upwards rather than exempted, at half the solver's recovery law, so the outward command passes; a stop must never trap the arm inside the margin it protects.

F. Postural clearance

The projection is minimal by construction, so when the frozen policy re-emits the same intent at every cycle the balance between intent and correction can freeze the arm on $h \approx 0$: safe, but not progressing. When the conflict is carried by an intermediate link we inject, *into the nominal velocity upstream of the QP*, a clearance term

$$\dot{\mathbf{q}}_{\text{clear}} = k_{\text{clear}}\, \rho(h)\, a\, \frac{\mathbf{N}\nabla h}{\|\mathbf{N}\nabla h\|}, \qquad \mathbf{N} = \mathbf{I} - \mathbf{J}^{+}\mathbf{J}, \qquad (18)$$

with a proximity ramp ρ and an authority ramp a that fades out when little of the gradient lies in the nullspace. Being an objective term, it is filtered by the active half-spaces exactly like the nominal command and can therefore never compromise safety; it buys liveness, not safety. When the controlled link itself is blocked, no heuristic unblocking is applied and the system enters a safe stop.

G. Scope of the guarantee

In continuous time the formulation gives forward invariance of the safe set of the *learned* barrier. Eight assumptions separate that statement from execution, each paired with a telemetry channel and a documented fallback; the five most consequential appear in Table II. The others are an initial state inside $\mathcal{C}$, handled by the recovery law of (9) which yields attractivity in place of invariance, the smoothness of the out-of-domain join, which occurs only beyond 40 mm of clearance, and gradient reliability, monitored through $\|\nabla h_l\|$ against $g_{\min}$. Two points deserve emphasis. The certificate

TABLE II: Monitored assumptions separating the continuous-time certificate from execution.

Assumption	Monitored quantity	Behaviour on violation
Constant selection	generation counter published with the state	recovery regime, capped repair absorbs the re-introduced point
Feasibility	`constr_final` on the published command	residual measured, not assumed; brake holds the command
Command tracking	projected perturbation on active lines	reflex veto $\alpha = 0$ until the measure returns below ε_{p}
Continuous time	1 kHz re-check of the held command	sampled tightening (12); command held if the brake fails
Static environment	per-link residual $\hat{\dot{d}}_l$	tightening η_l; robustification, no formal claim

applies to the learned barrier, not the true geometry, so physical clearance inherits the field approximation error, which is why Sec. VI-B reports the tail of that error and not only its mean.

Second, the filter constrains the exact value $h_{\min,l}$ while using the soft-min gradient, and the only function consistent in both is the compensated barrier $h_{\text{corr}} = h^{\text{soft}} + \alpha \log N$: it is the invariance of $\{h_{\text{corr}} \geq 0\}$ that the theorem delivers. Since $h_{\min,l} \leq h_{\text{corr}}$ and b is decreasing, the applied constraint is strictly stronger than the one h_{corr} would require: the implementation is more conservative than the certificate it invokes. The gap, bounded by $\alpha \log N \approx 8.3$ mm, is not margin consumed in operation but the distance between what the theorem guarantees and what the filter holds.

VI. Experiments

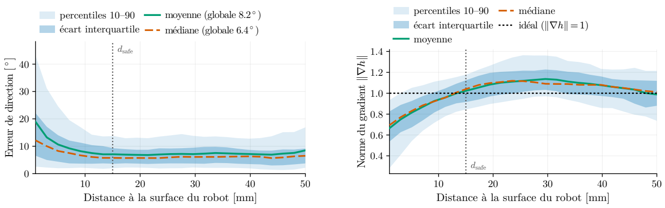
A. Setup

All experiments run on ROS 1 Noetic with Gazebo 11, on a 7-DoF Franka Emika Panda, with every learning component on a single NVIDIA RTX 3090 shared by all modules. State reading, inverse kinematics (Pinocchio [20]) and the reflex brake are C++; everything else is Python with PyTorch. Two tasks are used: assisted feeding with a fork and a wrist-mounted RealSense D415, and pick-and-place with the Franka gripper, trained on 60 and 40 demonstrations respectively, the simulated ones collected in ManiSkill 3 [21].

The distance field is evaluated against exact meshes by casting, per protected body, 40 rays along outward normals with 25 samples per ray between 1 and 50 mm. Only clean rays are kept, those whose exact distance to the union of bodies matches the advance along the normal, and samples outside a body's training domain are excluded (7.7 %) since the model there returns (4), leaving 5541 verified exterior points.

B. Distance and gradient accuracy

On value, the mean absolute distance error is 1.96 mm with a negligible bias of -0.02 mm and a correlation of $R = 0.9869$; the signed error distribution is centred and symmetric, and its 99th percentile in absolute value is 6.8 mm. That tail, not the mean, is the number the safety margin must face.



(a) (b)

Fig. 2: Gradient quality against exact meshes over 5541 verified exterior points. (a) Direction error, $5.4°$ median. (b) Gradient norm against the Eikonal value of one. Both are well conditioned at the margin d_{safe}. Labels in French.

The gradient is evaluated on the two properties the shield consumes. The median angular error against the exact direction is $5.4°$ and stays under $10°$ for most points beyond 5 mm from the surface (Fig. 2(a)); the norm approaches the Eikonal value of one beyond 10 mm and sags towards 0.7 very close to the surface, where the polynomial smooths the zero crossing (Fig. 2(b)). What matters is the behaviour at the margin: at d_{safe} the direction error is about $6°$ and the norm close to one, so the constraint pushes the arm the right way with the right amplitude at the precise distance where the filter activates.

Read against the margin, the filter constrains $h_{\min,l}$, so the learned distance to the nearest body is at least 15 mm in steady state, leaving at least 8.2 mm of physical clearance once the 99th-percentile model error is subtracted. In the formal worst case a single isolated obstacle point lowers that floor by $\alpha \log N \approx 8.3$ mm to 6.7 mm, comparable to the model tail; the two worst cases cannot be excluded jointly by computation alone, which is why physical clearance is validated empirically against exact meshes rather than deduced from the certificate.

C. Closed-form gradient against automatic differentiation

We compare the closed-form gradient (7) with `torch.autograd` on the joint evaluation of the barriers and the full $L \times 7$ constraint Jacobian for 100 obstacle points, over 100 repetitions. The maximum discrepancy between the two Jacobians stays at single-precision roundoff, from 6.0×10^{-8} at one link to 3.0×10^{-7} at eight, so (7) is exact and the speedup costs no accuracy; and the analytic form is systematically faster, from 5.60 against 7.18 ms at one link to 7.70 against 13.48 ms at eight, a ratio growing from $1.28\times$ to $1.75\times$ because automatic differentiation must retain and traverse a graph whose size grows with the chain while (7) is a fixed pair of contractions per point.

Joint evaluation of distance and gradient on GPU forms a plateau of about 5 ms up to a few thousand query points, dominated by fixed per-call overhead and growing only past roughly 4096 points, far beyond the operating point of $K = 64$; the exact mesh proximity query grows linearly and exceeds the 10 ms budget by one to two orders of magnitude, which rules it out online.

D. Lipschitz constants: a valid bound we could not use

The sampled-data tightening (12) and the reflex condition (17) both consume a per-link constant, and we report the negative result because it shaped the design.

The quantity the certificate needs is one-sided: the Taylor lower bound holds as soon as $L_l \geq \sup_{\boldsymbol{\xi}}(-\lambda_{\min}(\nabla^2_{\mathbf{q}} h_l(\boldsymbol{\xi})))_+$, since only negative curvature threatens it while positive curvature tightens it for free. The two-sided Lipschitz condition implies this but demands far more, and the distinction is not academic for a distance barrier, whose exact distance is convex outside a convex link with large curvature precisely on the harmless side. Because (2) is a Bernstein tensor product, its second derivatives are Bernstein polynomials whose coefficients are finite differences of the $w^{(l)}_{abc}$, and the convex-hull property bounds each Hessian entry by the largest of them; propagating through the normalisation and the kinematics yields a fully offline analytic bound. It is valid, no sample exceeds it, and it is unusable: 1.4×10^3 to 2.2×10^4 m/rad^2, roughly forty times the worst observed case, and since the recoil grows linearly in L_l it would freeze the arm long before it reached an obstacle. Two causes combine: the supremum covers the whole cube including corners the robot never visits, and coefficient arithmetic is two-sided, so it cannot isolate the negative curvature that alone matters.

The deployed constants come instead from a sampled campaign on the *deployed* gradient implementation: 8192 configurations drawn uniformly in the joint limits, one query point per link with a third of the draws outside the domain to exercise the extension branch, and the Hessian obtained by central differences of the double-precision analytic gradient. The hierarchy is strongly inhomogeneous, which justifies a vector L_l rather than one constant: body links peak between 2.2 and 25.9 m/rad^2, distal links between 44.3 and 79.3, and the fingers dominate by an order of magnitude at 508. That finger excess is not geometric, a finger being convex, but comes from polynomial oscillation around sharp edges; retraining both finger models at $n = 12$ halved their worst case from 840 to 508 while improving value and direction accuracy. We deploy the 99th percentile rather than the supremum, from 1.0 for link 2 to 32.0 for link 7, with a further reduction to 20 for the fingers, and monitor the hypothesis online through $\hat{L} = \|\Delta \nabla h_l\| / \|\Delta \mathbf{q}_0\|$ differenced between consecutive solver states of one selection generation.

Two limits of validity are stated rather than hidden. The out-of-domain join is only piecewise differentiable, so no finite constant bounds the gradient variation across it, though those joins occur only beyond 40 mm of clearance where the certificate holds ample reserve. And the soft-min contributes $-\frac{1}{\alpha}(\sum_i \omega_i \mathbf{g}_i \mathbf{g}_i^\top - \bar{\mathbf{g}}\bar{\mathbf{g}}^\top)$ to the Hessian, negative semi-definite and therefore on the threatening side, which the single-point campaign does not measure. It remains visible to the online estimator, so the mechanism is runtime assurance, not a proof.

E. Real-time budget of the integrated system

Table III reports per-stage timings measured in closed loop with all modules sharing the GPU. Three readings matter.

The reflex tick holds a hard deadline: 1.00 ms period with zero standard deviation over 9013 consecutive ticks, and the computation it hosts consumes 0.01 ms median and 0.03 ms at p95, 1 to 3 % of its period. This is the layer on which

TABLE III: Per-stage timings of the reactive layers in closed loop (ms). Start-up cycles, where CUDA graph capture and first allocation accumulate, are excluded.

Stage	**Role**	**n**	**Mean**	**Med.**	**p95**
Reflex brake, 1 kHz					
`reflex_period`	real-time tick	9013	1.00	1.00	1.00
`reflex_compute`	factor α	9012	0.02	0.01	0.03
SDF-CBF shield, 100 Hz					
`point_selection`	top-K selection	5077	1.03	0.44	3.34
`projection`	Dykstra sweeps	819	1.52	1.33	2.68
`cbf_correction`	correction cycle	2554	1.95	0.98	7.14
`cbf_command`	command publish	2554	0.04	0.01	0.04

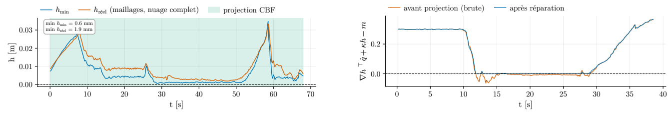
the continuous-time assumption of Table II rests, and its slack is such that contention on the shared GPU, which does affect the Python stages, cannot compromise it. The dominant shield stage runs at 0.98 ms median and 7.14 ms at p95 under a 10 ms period, and critical point selection at 0.44 ms median, more than twenty times faster than the 10 ms staleness provisioned in the margin budget. Dykstra sweeps are invoked on only 819 of 2554 correction cycles, so the solver is exercised on 32 % of cycles, those with at least one active constraint: that ratio measures the shield's transparency directly and explains the median-to-p95 spread, whose tail is constrained cycles rather than contention.

Finally, replanning is limited by tracking, not by compute. ODE integration costs 4.79 ms median, and the whole chain from conditioning encode to timestamped joint trajectory about 40 ms on average and 83 ms at p95, well under the 200 ms available at 5 Hz. The internal split is counter-intuitive: converting the model output into an executable joint trajectory (20.70 ms median) costs more than four times the ODE integration, and inverse kinematics over the 16 waypoints (5.12 ms) costs more than generation. The generative model is the cheapest stage of its own chain.

F. Calibrating the monitored tracking bound

The tightening (11) rests on a bound ε_{p} on the adverse projected perturbation, calibrated on 26 independent pick-and-place trajectories with varied planner seeds and a 45 s cap, all kept (19 completions, 7 timeouts) since safety calibration must not be conditioned on task success. Each contributes, at 100 Hz, the worst 1 kHz sample per 10 ms window of $-\nabla h_l^\top \mathbf{d}/\|\nabla h_l\|$ on active lines, split by brake regime.

Over the 31 094 effective-execution windows ($\alpha > 0.9$) the median is 0.0005 rad/s and the 99th percentile 0.032 rad/s, so the deployed $\varepsilon_{\mathrm{p}} = 0.025$ rad/s covers 97.9 % of windows, a fraction that upper bounds the excess rate over individual ticks since each window keeps its worst tick. We are explicit about what this value is: ε_{p} is not a strict majorant but the monitor trigger threshold, placed at the 98th percentile. The tightening certifies the realised flow on the mass of the distribution, per active line, while each residual excursion is detected online and handled by the reflex veto; safety rests on that mass-and-tail partition plus the d_{safe} reserve, not on ε_{p} as a ceiling. Braking windows, excluded from the veto criterion since the brake's own certificate covers them, show a structurally larger discrepancy (10.4 % above ε_{p}).



(a) (b)

Fig. 3: Closed-loop run. (a) Exact barrier $h_{\mathrm{min},l}$, projection-active and repair-only cycles shaded. (b) Safety constraint on the nominal velocity against the published command. Labels in French.

The cost of the alternative is quantified at trajectory level. Treating the 26 per-trajectory maxima as exchangeable with that of a future trajectory, the order statistic gives a distribution-free prediction bound [22]: with probability at least $26/27 \approx 0.963$, the worst excursion of a new trajectory does not exceed 0.092 rad/s in effective execution, or 0.173 rad/s including braking. Those are the values a static unmonitored majorant would have to load into (11) on every active line, 3.7 to 6.9 times the deployed bound, and with them the standoff distance it imposes near the boundary. The monitored design escapes that quantified cost.

G. Closed-loop behaviour

Fig. 3(a) follows the exact barrier $h_{\mathrm{min},l}$ over a representative feeding run with obstacles, marking separately the cycles where the projection corrects the velocity and those where only the repair step intervenes. The regime is visible directly: after the approach phase the system settles into boundary following, the barrier stabilising a few millimetres above zero while the robot advances tangentially along the surfaces towards the target, which is exactly what a projection should do. The barrier stays non-negative throughout the run.

Fig. 3(b) contrasts the safety constraint evaluated on the nominal velocity with the same constraint evaluated on the published command. The nominal curve dives below zero as soon as the robot operates near obstacles, down to -0.087: left alone, the Flow Matching trajectory would violate the safety condition continuously, which is the quantitative statement of the problem this paper addresses. The published curve never falls below -10^{-5}, the floating-point rounding floor, over the whole run. The repair step is far from marginal in that result, firing on nearly every cycle where the projection is active because the output smoothing systematically erodes part of the correction within the same cycle. This is the empirical content of re-measuring admissibility on the published command instead of trusting solver convergence.

The chain from command to execution closes as well: the commanded barrier rate $\nabla h^\top \dot{\mathbf{q}}_{\mathrm{final}}$ and the realised one $\nabla h^\top \dot{\mathbf{q}}_{\mathrm{mes}}$ track each other, so the corrections the filter admits are actually realised by the low-level controller. That property is not automatic. Two position-based interfaces evaluated during development failed symmetrically: re-anchoring the reference on the measured position every cycle drove the realised-to-commanded ratio to zero, corrections being reabsorbed before being realised, while integrating it at a step larger than the true loop period drove the ratio negative and above one in magnitude, the arm crossing the boundary in oscillation. Passing the certified velocity directly to a joint velocity controller removes both regimes by removing the intermediary. Consistently with the calibration above, the 99th percentile of the projected perturbation over effective-execution cycles of this run, 0.0247 rad/s, sits right at the deployed bound.

VII. Limitations and Conclusion

The guarantee is about safety, not task completion. Postural clearance extends liveness only to conflicts carried by the arm body and resolvable in the task nullspace; when the obstacle blocks the controlled link itself, the projection cancels the incoming velocity component, the frozen policy re-emits the same intent at every replanning cycle, and the system stays indefinitely in a safe stop, the barrier positive but the task stalled. Resolving topological dead ends belongs to the nominal planning layer.

The margin is sized for simulation, where obstacle clouds are sampled on exact surfaces and frame transforms are known, so hardware deployment would require enlarging d_{safe} by the perceptual error budget. The certificate applies to the learned barrier, not the true geometry, and is quasi-static: η_l robustifies against environment motion with no formal claim against moving obstacles. Closed-loop replanning also loses a measurable fraction of the commanded speed to vector cancellation, the nominal direction jumping at every regeneration while the low-pass filter averages vectors whose direction alternates; temporal aggregation of successive generations is the natural remedy, left to future work with the statistical A/B campaign and cross-embodiment transfer.

We presented a manipulation system in which task intent and safety are decoupled end to end: a frozen Flow Matching policy proposes, and a whole-body CBF shield backed by a 1 kHz reflex brake disposes. The enabling component is a Bernstein robot distance field refined for gradient fidelity against verified normal rays, paired with an exact closed-form joint-space gradient and a monotone out-of-domain minorant. The distinguishing feature of the safety layer is not any single tightening but the discipline applied to all of them: every gap between the continuous-time theorem and sampled execution is named, tightened, and given a telemetry channel, and the constants they consume are calibrated on measured distributions with distribution-free coverage rather than asserted as global majorants. Future work targets hardware deployment with a perceptual error budget and a second safety level based on external force estimation, for the intentional contact that assisted feeding ultimately requires.

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